Phishing Detection and Awareness Report

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Introduction

What is Phishing

Phishing is a type of cyber attack in which attackers send fake emails or messages to trick users into revealing sensitive information such as passwords, personal details, or financial data. These emails often appear to come from trusted sources like banks, companies, or government organizations.

Why it is Dangerous

Phishing is dangerous because it targets human behavior rather than technical systems. Users may unknowingly share confidential information, which can lead to identity theft, financial loss, or unauthorized access to accounts. Since phishing emails often look legitimate, they can easily deceive users who are not aware of such attacks.

What this Report Does

This report analyzes a phishing email sample to identify common phishing indicators such as fake identity, suspicious sender details, and social engineering tactics. It also includes email header analysis using MXToolbox, classifies the risk level, explains how the attack works, and provides guidelines to help users recognize and avoid phishing attacks.

Phishing Email Sample

Subject: Attention: I am Diplomatic agent Mr. Wood Forest

From: Mr. Wood Forest <nemani.tukunia@postfiji.com.fj>

Email Content:

Attention??

I am Diplomatic agent Mr. Wood Forest, I have just arrived here in
your country with your consignment box of funds sum of $10.5 million USD.

I'm here at the airport right now and I need you to reconfirm your
home address and current cell phone so that I can deliver the package.

Please respond urgently as I do not want to deliver to the wrong destination.

Thanks,
Diplomatic agent Mr. Wood Forest
IMF Agent
Contact Email: mywoodforestbiz.7@gmail.com

This email is take from a sample phishing dataset provided on GitHub

Phishing Indicators (Analysis)

- **Fake Identity:** The sender claims to be a diplomatic agent without verification.
- **Urgency Tactic:** The email pressures the user to act quickly since the attackers want users to act without thinking.
- **Lack of Professionalism:** The email contains grammatical errors and informal tone.
- **Logical Inconsistency:** A real delivery agent would already have recipient details.
- **Too Good to Be True:** The sum amount is unusually large which is commonly used in scam emails.

Email Header Analysis (Tool-Based)

- **MXToolbox was used for analyzing the email header.**

- **SPF Failure:** SPF (Sender Policy Framework) failure indicates that the sender is not authorized to send emails from that domain, suggesting possible spoofing.

- **Reply –To vs From:** The sender uses an email address from postfiji.com while Reply-To address is a Gmail account which indicates they want replies are redirected to a different , potentially malicious account.

- **Unrelated Sending Server:** The email originates from a server (Russian-based server) and IP address unrelated to the claimed identity (Diplomatic agent/Fiji), indicating that the message is sent from an untrusted source.

- **DKIM Missing:** DKIM is not present, which means that the email lacks a digital signature and its authenticity cannot be verified.

- **DMARC Missing:** DMARC is not configured, indicating a lack of protection against email spoofing and phishing attempts.

Headers Found

Header Name	Header Value
Authentication-Results	spf=fail (sender IP is 109.202.24.52) smtp.mailfrom=postfiji.com.fj; dkim=none (message not signed) header.d=none;dmarc=none action=none header.from=postfiji.com.fj;compauth=fail reason=001
Received-SPF	Fail (protection.outlook.com: domain of postfiji.com.fj does not designate 109.202.24.52 as permitted sender) receiver=protection.outlook.com; client-ip=109.202.24.52; helo=54upr.rosreestr.ru;
X-IncomingTopHeaderMarker	OriginalChecksum:967488A6B56FE8D6811701FA86525B8497BBF9854ED315EE4582192C07F4278A;UpperCasedChecksum:D916BCFBE40B788B40930A4A6CECD75BBB5EB59F8050234DBC4BE9F69B309D7A;SizeAsReceived:734;Count:13
Content-Type	text/plain; charset="iso-8859-1"
Content-Transfer-Encoding	quoted-printable
Content-Description	Mail message body
Subject	Attention:I am Diplomatic agent Mr.Wood Forest,
To	Recipients <nemani.tukunia@postfiji.com.fj>
From	"Mr.Wood Forest" <nemani.tukunia@postfiji.com.fj>
Date	Sat, 29 Jul 2023 13:03:39 -0700
Reply-To	mywoodforestbiz.7@gmail.com
Return-Path	nemani.tukunia@postfiji.com.fj
Message-ID	<CENTRALoY5uIBUevIXn0001393a@54upr.rosreestr.ru>
X-OriginalArrivalTime	29 Jul 2023 20:00:51.0636 (UTC) FILETIME=[6037E740:01D9C257]
X-IncomingHeaderCount	13

Figure: Email header analysis using MXToolbox showing SPF failure, missing DKIM, and suspicious sender details.

Risk Classification

- **Classification: Phishing**

- **Reasons:**
  - The email uses a fake identity and unrealistic offer
  - It attempts to collect personal information from the user.
  - Header analysis shows spoofing, indicated by SPF failure and mismatched sender details.

How the Attack Works

- **Step 1: Email Arrival**
  User receives the email from ‘Diplomatic Agent’ and may trust it due to the authority of the sender and the promise of a large sum of money.

- **Step 2: User is Manipulated**
  The email creates excitement to the user and a sense of urgency, using greed and pressure to influence the user’s decision without careful thinking.

- **Step 3: User Shares information**
  The email asks for personal details such as address and phone number, making the user share sensitive information.

- **Step 4: Attacker Benefits**
  The attacker can use this information to carry out further scams, contact the user or request money under false pretenses.

Prevention Guidelines

Do’s

- Verify the sender’s email address before trusting the message
- Be cautious of emails offering large sums of money or unexpected rewards
- Check for inconsistencies in the message content and sender details
- Report suspicious emails to the organization’s security team

Don’ts

- Do not share personal or sensitive information through email
- Do not trust emails that create urgency or pressure you to act quickly
- Do not reply to suspicious emails or unknown senders
- Do not click on links or download attachments from untrusted sources

Conclusion

Phishing is a serious cyber threat that targets users by exploiting trust and human behavior rather than technical vulnerabilities. It can lead to data theft, financial loss, and unauthorized access to sensitive information. Therefore, it is important for users to stay aware and cautious while handling emails, especially those from unknown or suspicious sources. By recognizing common phishing indicators and following basic security practices, users can effectively prevent such attacks. Awareness and vigilance are the key factors in protecting individuals and organizations from phishing threats.